Short-term technical preparation

Kit and equipment:
Selection, environmental and cultural factors, effects on performance

Ergogenic aids for short-term preparation:
Drugs/supplements, clothing (compression clothing, ice vests), cycle ergo meters for warm-up

Use of holding camps and pre-match rituals

- The differences between short-term holding camps and long-term preparation camps
- Use of pre-match rituals in preparation for optimum performance

Fatigue and the recovery process

Fatigue:
Effects on performance, depletion of fuels (PC, glycogen), waste product accumulation (lactic acid), central governor theory, dehydration, electrolyte loss

Recovery:
Time phases (first few hours — 24 hours)

- First few hours — cooling down, lactic acid removal, restoration of ATP-PC and glycogen stores, EPOC (excess post-exercise oxygen consumption)
- 24 hours — DOMS (delayed onset muscle soreness), hydration, carbohydrate loading

Use of ergogenic aids — ice baths, compression clothing, music

3.2 Long-term Preparation

Long-term physiological preparation

Key long-term adaptations linked to training methods:
Students need to understand periodisation, and other long-term strategies, and be able to identify how they enable structural and functional adaptations to be planned for, and achieved, including:

- Aerobic training — increases in stroke volume, cardiac output, number of capillaries, mitochondria, reduction in body fat/ increase in lean body mass, increased lactate threshold, VO_2 max
- Anaerobic training — increases in PC stores, more anaerobic enzymes, greater lactic acid tolerance
- Continuous training — improves aerobic system (see above), improved use of fat for energy, improved lactate threshold
- Interval training — can be aerobic or anaerobic depending on structure — see above
- Plyometrics/power training — increases in rate of fibre recruitment, number of fibres recruited, rate of force production/ speed, improved co-ordination
- Circuit/weight/resistance training — increases in size of muscle fibres, muscle mass, lean body mass, force production
- Speed training — improved acceleration, sprinting speed (usually associated with increased fibre recruitment/number of fibres recruited)
- Fartlek training — combination of aerobic, anaerobic, speed, interval and continuous
- Core stability training — improvements in balance, co-ordination
- Speed, Agility, Quickness (SAQ) training — improved speed, agility, reactions and co-ordination
- Stretching (static, ballistic, dynamic, PNF) — increased RoM

Long-term psychological preparation

Goal setting and mental training:
SMART (specific, measurable, achievable, realistic, time-bound) targets, performance profiling, attribution theory

Motivation:
Achievement motivation, nAch/nAff/Intrinsic/Extrinsic Attribution theories